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the past five months, both Jensen Huang, CEO of the American computer chipmaker Nvidia, and Sam Altman, who heads OpenAI, have visited India, emphasising the country's growing significance in the tech world, including in the field of Artificial Intelligence (AI). Nvidia reports a tenfold increase in the deployment of its graphics processing units (GPUs)—specialised chips used to train AI models—in India over 18 months, while for OpenAI, India has emerged as the second-largest market, with users tripling in the past year.

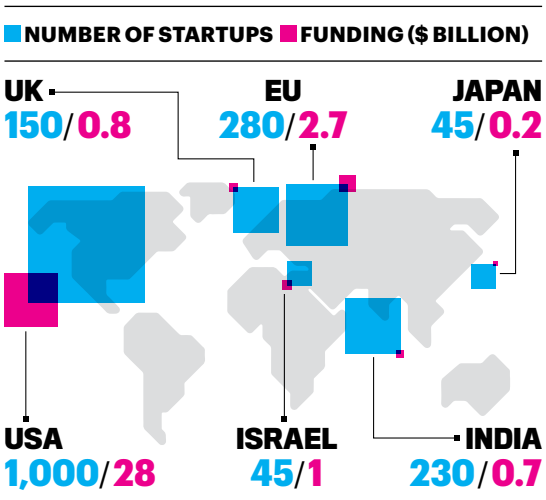
During his February 5 visit, Altman said that India should be "one of the leaders of the AI revolution", an endorsement in stark contrast to his 2023 stance that the country "should not even try" to build foundational AI models. What has changed in between, of course, is the end-January release of industry-disruptive DeepSeek-R1, a large language model (LLM) developed by a Chinese startup based in Hangzhou.

Foundational LLMs are deep learning systems trained on gigantic amounts of data to perform tasks such as natural language processing, question-answering and image classification. They form the foundation on which conversational agents and other Generative AI (or GenAI) products are built. If prohibitive costs were what prompted some experts to advise caution in committing billions of dollars to the development of LLMs such as OpenAI's GPT-4, the arrival of DeepSeek has been a game-changer because the Chinese startup claims to have developed its eponymous model at a fraction of the cost incurred by its western forerunners.

"It's totally breaking the AI monopoly. There's a realisation that AI innovation is no longer the exclusive domain of a few tech companies," says Rajesh Nambiar, president of the Indian IT industry association Nasscom. "This whole DeepSeek moment is both an opportunity and a wake-up call for India in some sense. Basically, it reinforces the need for self-reliance in AI infrastructure, talent development and certainly open-source collaboration."

THE GenAI ECOSYSTEM

India is making its presence felt among the top global Generative AI ecosystems



Figures since 2023, are indicative; Source: India's Generative AI Startup Landscape 2024, Nasscom

HOMEGROWN MODELS

On February 11, at the AI Action Summit in Paris, Prime Minister Narendra Modi touched upon all these developments as he announced India's plan to develop "its own LLM considering our [linguistic] diversity". In fact, there are several parallel efforts ongoing in this domain, including a foundational model funded by the Department of Science and Technology (DST). Called BharatGen, this suite of LLMs is being developed by a Technology Innovation Hub at the Indian Institute of Technology (IIT) Bombay. "By enabling an ecosystem around its foundational models, BharatGen envisions positioning India as a global leader in Generative AI," Prof. Abhay Karandikar, secretary, DST, tells INDIA TODAY. "BharatGen will work with the ministry of electronics and information technology (MeitY) and its IndiaAI Mission."

A six-month-old venture with a funding commitment of around Rs 236 crore over two years, BharatGen has been incorporated as a non-profit organisation. Being built from the ground up, the model spans text-to-speech and automatic speech recognition in multiple Indian languages. In this, it leveraged the work done by Bhashini, a natural language processing project for Indian languages under the MeitY. "We have also been building homegrown datasets. As Indian languages have less than one per cent representation on the internet, we need to curate original datasets," says Prof. Ganesh Ramakrishnan of IIT Bombay, who leads the BharatGen consortium. "Not only is BharatGen government-funded, it is an academic collaboration by design."

As of now, BharatGen has developed a text mod-